Estimation & Detection Theory

TAMU ECEN-662

Brody Jordan

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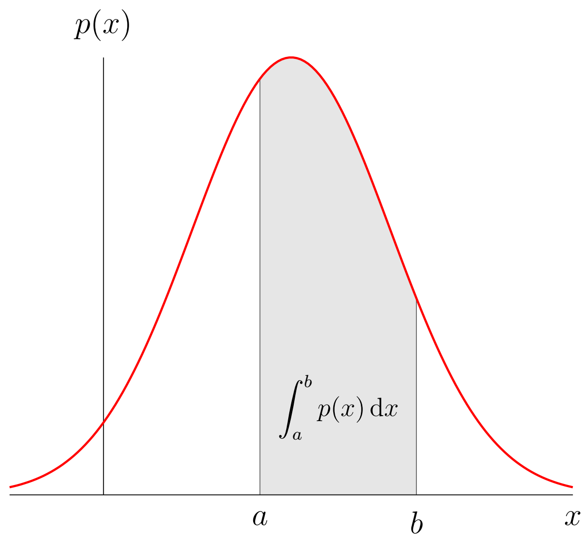

Acknowledgements

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Contents

Chapter 1

Overview

Suppose that the distribution of random variable $X \in Xc$, is determined by a parameter $\theta \in \Theta$:

$$X \sim f_\theta(x)$$

Here, the functional form of f_θ is known, but the value of θ is unknown. We use f_θ to denote probability density function (pdf) and p_θ to denote probability mass function (pmf), and f_θ for generic descriptions.

In statistical inference, we observe a realization x of X and need to answer some questions about θ, such as:

- Estimation: what is a good guess of θ?
- Detection: is $\theta \in \Theta_1$, or is $\theta \in \Theta_2$?
- a
- b
- c

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$p_\theta(x_i) = \theta^{x_i}(1-\theta)^{1-x_i}$ piecewise trick.

$p_\theta = \frac{1}{\binom{n}{t}}$ not dependent on x **Jan. 15**

Model $x \propto f(x)$ distribution unknown, parameterized by unknown θ. By looking at x, make inference θ. Dependency of x on θ is only on the distribution.

Statistic: $T = T(x)$ function of random variable; does not depend on θ.

T $f_\theta(t)$ potentially not as accurate as the inference made from x. When mapping T is invertible, there is no difference between x and t.

If information is sacrificed (T loses information on θ).

Chapter 2

Sufficiency

Sufficient statistic;

In case where mapping does not lose any information, that statistic is called a 'sufficient' statistic. Invertible mapping preserves all data.

Want sufficient statistic that compresses data and preserves information...

Sufficient: If conditional distribution does not depend on θ, $f_\theta(x|t)$ $p_\theta(x|t) = \frac{p_\theta(x)}{p_\theta(t)}$

Fisher-Neyman factorization theorem: allows us to identify sufficient statistics direclty from the likelihood function $f_\theta(x)$. can be taken as a working definion of sufficiency THM: let $f_\theta(x)$ be a pdf or pmf of X. Stat $T = T(X)$ is sufficient for θ iff $\exists$ $g_\theta(t)$ and $h(x)$ s.t.: $f_\theta(x) = g_\theta(T(x))h(x)$ where $h(x)$ does NOT depend on θ

Bernoulli-trials revisited: PMF of X $p_\theta(x) = \theta^{T(x)}(1-\theta)^{n-T(x)}$ here x only shows up as part of $T(x)$ therefore by the Fisher-Neyman thm, $T = \sum_{i=1}^n X_i$ is a sufficient statistic for θ.

X is discrete, so

$$p_\theta = \sum_{y:T(y)=t} p_\theta(y)$$

thus for any (x, t) such that $T(x) = t$, we have

$$p_\theta(x|t) = \frac{h(x)}{\sum_{y:T(y)=t} h(y)}$$

now ratio does not depend on θ, so by definition $T = T(x)$ is a sufficient statistic for θ.

If $p_\theta(x) = h(x)g_\theta(T(x))$...

Example: Gaussian with unknown mean. Suppose we observe $X = [X_1, \ldots, X_n]^t$, where

$$X_i{}^{iid}\mathcal{N}(\mu, \sigma^2)$$

where σ^2 is a known constant and μ unknown parameter

We have $f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$ and hence,

$f_\mu = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{\mu}{\ }...\right) = g_\mu(T(x)h(x)$

Example: both μ and σ^2 unknown and $\theta = [\mu, \sigma^2]^t$ is a two-dimensional vector

We have $f_\theta(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp(...)$

with $T(x) = [\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2]$

Example: Uniform with unknown parameter

$$X_i{}^{iid}\mathcal{U}(0, \theta)$$

with $\theta > 0$ unknown parameter, want to make inference on upper-bound

We have

$$f_\theta(x_i) = \frac{1}{\theta} 1_{x_i \leq \theta}$$

for any $x_i \geq 0$, hence

$$f_\theta(x) = \frac{1}{\theta^n} \Pi_{i=1}^n 1_{x_i \leq \theta} = \frac{1}{\theta^n} 1_{\max_{i \in [n]} x_i \leq \theta}$$

Therefore $T(X) = \max_{i \in [n]} X_i$ is a sufficient statistic

Chapter 3

Minimal Sufficiency

A sufficient statistc is minial if it is a function of any other sufficient statistics

3.0.1 Certification of minimality

The minimality of a sufficient statistic can be verified based on the following proposition: a sufficient statistic T is minimal if for any $x, y \in \mathcal{X}$,

$$\frac{f_\theta(x)}{f_\theta(y)}\text{is independent of } \theta \Rightarrow T(x) = T(y)$$

if the ratio of their likelihoods is independent of θ, then need to check if it implies $T(x) = T(y)$. Only a sufficient condition.

Note that

$$\frac{f_\theta(x)}{f_\theta(y)} = \left(\frac{\theta}{1-\theta}\right)^{T(x)-T(y)}$$

Clearly, the ratio does not depend on θ iff $T(x) = T(y)$. **Jan. 20**

Any function of the observation that preserves all the information is a sufficient statistic.

Minimal sufficient statistic

A sufficient statistic is *minimal* if it is a function of any other sufficient statistics. This is the "optimal" data compression.

A sufficient condition for minimality given sufficiency: The minimality of a sufficient statistic can be verified based on the following proposition: A sufficient statistic T is minimal for any $x, y \in \mathcal{X}$,

$$\frac{f_\theta(x)}{f_\theta(y)} \text{ is independent of} \theta \Rightarrow T(x) = T(y)$$

(Proof) Let T be a sufficient statistic that staisfies the above property. To show that T is minimal, we must show that for any other statistic T', T is a function of T'.

Now suppose that $T'(x) = T'(y)$. We have

$$\frac{f_\theta(x)}{f_\theta(y)} = \frac{g'_\theta(T'(x))h'(x)}{g'_\theta(T'(y))h'(y)} = \frac{h'(x)}{h'(y)}$$

which is independent of θ

Previously, we assume T is a sufficient statistic and then provides a sufficient confition for the minimality.

Let $T = T(X)$ be a stat. It can be shown that T is a sufficient statistic if for any $x, y \in \mathcal{X}$,

$$T(x) = T(y) \Rightarrow \frac{f_\theta(x)}{f_\theta(y)} \text{ is independent of } \theta.$$

Proof based on Fisher-Neyman...

Combined with the previous proposition, a statistic $T = T(X)$ is a minimal sufficient statistic if for any $x, y, \in \mathcal{X}$,

$$\frac{f_\theta(x)}{f_\theta(y)} \text{is independent of} \theta \leftrightarrow T(x) = T(y)$$

Example 3.0.1 (Bernoulli trials)**.** Consider the statistic

$$T = \sum_{i=1}^{n} X_i$$

Note that

$$\frac{f_\theta(x)}{f_\theta(y)} = \left(\frac{\theta}{1-\theta}\right)^{T(x)-T(y)}$$

Clearly, the ratio $\frac{f_\theta(x)}{f_\theta(y)}$ does not dpeend on θ iff $T(x) = T(y)$.

Complete sufficient statistic

A statistic $T = T(X)$ is complete if for any real-valued function ϕ,

$$\mathbb{E}_\theta[\phi(T)] = 0, \forall \theta \in \Theta \Rightarrow \mathbb{P}_\theta(\phi(T) = 0) = 0, \forall 0 \in \Theta$$

Example 3.0.2 (Bernoullie trials)**.** Recall

$$T = \sum_{i=1}^{n} X_i$$

is a sufficient statistic for θ.

If for some real-valued function ϕ we have

$$\mathbb{E}_\theta[\phi(T)] = \sum_{t=0}^{n} \phi(t) \binom{n}{t} \theta^t (1-\theta)^{n-t} = 0$$

we must have

$$\sum_{t=0}^{n} \phi(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t = 0$$

for all $\theta \in (0, 1)$.

Example 3.0.3 (Uniform with unknwon support)**.** Recall that for uniform random variables over an unknown support $[0, \theta]$.

$$T = \max_{i \in [n]} X_i$$

is a sufficient statistic for θ. The pdf of T can be calculated as

$$f_\theta(t) = nt^{n-1}\theta^{-n}1_{0<t\leq\theta}$$

If for some real-valued function ϕ we have

$$\mathbb{E}_\theta[\phi(T)] = n\theta^{-n}\int_0^\theta \phi(t)t^{n-1}dt = 0$$

for all $\theta > 0$, we must have

$$g(\theta) = \int_0^\theta \phi(t)t^{n-1}dt = 0, \quad \forall\theta > 0$$

Taking the derivative, $g'(\theta) = \phi(\theta)\theta^{n-1}$.

] **Jan. 22**

3.1 Complete sufficient statistic

Under mild conditions, it can be shown that if a sufficient statistic is complete, it is also minimal.

Let T be a complete sufficient statistic and T' be another sufficient statistic. We need to show that T is a function of T'.

Let

$$\begin{aligned}\psi(t') &:= \mathbb{E}_\theta[T|T' = t'] \\ \rho(t) &:= \mathbb{E}_\theta[\psi(T')|T = t]\end{aligned}$$

By law of total expectation,

$$\begin{aligned}\mathbb{E}_\theta &= \mathbb{E}_\theta[\mathbb{E}_\theta[T|T']] \\ &= \mathbb{E}_\theta[\psi(T')] \\ &= \mathbb{E}_\theta[\mathbb{E}_\theta[\psi(T')|T]] \\ &= \mathbb{E}_\theta[\rho(T)]\end{aligned}$$

for any $\theta \in \Theta$. By the completeness of T, we have $\mathbb{P}_\theta(T = \rho(T)) = 1$.

Recall the law of total variance: $\mathrm{Var}[Y] = \mathbb{E}[\mathrm{Var}[Y|Z]] + \mathrm{Var}[\mathbb{E}[Y|Z]]$. Now we have

$$\mathrm{Var}_\theta[\rho_i(T)] = \mathbb{E}_\theta[\mathrm{Var}_\theta[\rho_i(T)|T']] + \mathbb{E}_\theta[\mathrm{Var}_\theta[\psi_i(T')|T]] + ...$$

Chapter 4

Exponential Family of Distributions

Jan. 22

4.1 Lecture 3: Exponential Family of Distributions

A model $f_\theta(x)$ is called an exponential family of distributions if it can be written as

$$f_\theta(x) = h(x)\exp(\eta^t T - A(\eta))$$

where $\eta = \eta(\theta)$ is an s-dimensional function of the model parameter θ; $T = T(x)$ is an s-dimensional function of the observation x; $h(x)$ is a one-dimensional function of the observation x known as the base measure; $A(\eta)$ is the normalization factor known as the log-partition function and can be computed as:

$$A(\eta) = \log\left(\int_{\mathcal{X}} h(x)\exp(\eta^t T(x))\,dx\right)$$

In addition, it is required that the support set $\{x : f_\theta(x) > 0\}$ does not depend on θ. Note that the likelihood function f_θ depends on the model parameter θ only through $\eta = \eta(\theta)$. Therefore, η is known as the natural parameter of the model, and any representation of the aforementioned form is known as canonical representation of the expoential family. Note that for any given expoential family of distributions, canonical presentations are not unique. For example, if $(\eta, T, h, A(\eta))$ is a canonical representation, then

$$(\eta' = B\eta,\, T' = B^{-1}T,\, h' = h,\, A'(\eta') = A(B^{-1}\eta'))$$

is also a canonical representation for any invertible matrix B. A canonical representation is called irreducible if neither $(\eta_i(\theta) : i \in [s])$ nor $(T_i(x) : x \in [s])$ satisfies a linear-equality constraint; otherwise, we can find an alternative canonical representation with a natural parameter of a reduced dimension.

4.1.1 Full-ranked and curved families

Without loss generality, we shall assume all canonical representations of an expoential family are irreducible. Let Θ be the space of the model parameter θ and let $\eta(\Theta) := \{\eta(\theta) : \theta \in \Theta\}$ be the space of the natural parameter η induced by $\eta = \eta(\theta)$.

We say that an (irreducible) s-parameter expoential family is full-ranked if the space of the natural parameter $\eta(\Theta)$ contains an s-dimensional rectangle; otherwise, it is called a curved family. Note that by the Fisher-Neyman factorization theorem, $T(X)$ is a sufficient statistic for θ for any expoential family. In addition, if the expoential family is full-ranked then $T(X)$ is also complete for θ.

Multiple independent measurements

Let $X = [X_1, \ldots, X_n]^t$, where

$$X_i \overset{iid}{\sim} f_\theta(x_i)$$

and

$$f_\theta(x_i) = h(x_i)\exp(\eta^t T(x_i) - A(\eta))$$

we thus have

$$f_\theta(x) = \prod_{i=1}^n f_\theta(x_i) = \left(\prod_{i=1} h(x_i)\right)\exp\left(\eta^t \sum_{i=1} T(x_i) - \eta A(\eta)\right)$$

For Bernoulli trials, we have

$$\begin{aligned} p_\theta(x_i) &= \theta^{x_i}(1-\theta)^{1-x_i} \\ &= \exp\left(x_i \log\frac{\theta}{1-\theta} - \log\frac{1}{1-\theta}\right) \end{aligned}$$

Thus,

$$p_\theta(x) = \prod_{i=1}^n p_\theta(x_i)$$

is also a one-parameter expoential family of distributions with

$$\eta(\theta) = \log\frac{\theta}{1-\theta}, \quad T(x) = \sum i = 1^n x_i, \quad h(x) = 1, \quad \text{and} \quad A(\eta) = n\log(1+e^\eta)$$

Note that the space of the model parameter θ is $(0, 1)$, the space space of the natural parameter is $\mathbb{R}_{++}$. Therefore, this expoential family is full-ranked, and the sufficient statistic $T(X) = \sum_{i=1}^n X_i$.

Example 4.1.1. For Gaussian with unknown mean, we have

$$f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{x_i^2}{2\sigma^2}\right)\exp(\ldots)$$

Thus,

$$f_\mu(x) = \prod_{i=1}^n f_\mu(x_i)$$

is also a one-parameter expoential family of distributions with

...

Example 4.1.2. For Gaussian with unknown mean and variance, we have

$$f_\theta(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(\frac{\mu x_i}{\sigma^2} - \frac{x_i^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2}\right)$$

which is a two-paremeter exponential family of distributions. ... with

$$\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right), \quad T(x) = \left(\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2\right), \quad h(x) = \left(\frac{1}{\sqrt{\pi}}\right)^n$$

Example 4.1.3. $X_i \sim \mathcal{N}$ Also a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right), T(x) = \left(\sum_{i=1}^{n} x_i, \sum_{x=1}^{n} x_i^2\right), h(x) = \left(\frac{1}{\sqrt{2\pi}}\right)^n$$

and log-partition function

$$A(\eta) = \frac{n}{2} - \frac{n}{2}\log(-\eta 2)$$

As discussed, in this case the two-dimensional sufficient statistic

$$T(X) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1} X_i^2\right)$$

is not complete for any $n \geq 1$.

Example 4.1.4. Uniform with unknown support. Recall that for each $i \in [n]$, the likelihood function

$$f_\theta(x_i) = \frac{1}{\theta} 1_{\{x_i \leq 0\}}$$

If the support depends on θ, then it is not an exponential family. In this case, the sum is not a sufficient statistic, but the max is.

$$T(X) = \max_{i \in [n]} X_i$$

rather than

$$T(X) = \sum_{i=1}^{n} X_i$$

Distributions and moments of $T(X)$

Chapter 5

UMVUE I

5.1 Gaussian with unknown mean and variance

The likelihood function is an exponential family of distributions with a complete sufficient statistic $T = (T_1, T_2)$, where

$$T_1 = \sum_{i=1}^{n} X_i \text{ and } T_2 = \sum_{i=1}^{n} X_i^2$$

Thus by the Lehmann-Scheffe theorem, any unbiased estimators that are a function of $T = (T_1, T_2)$ are UMVUE for the unknown parameters μ and σ^2.

Therefore, the empirical mean

$$\hat{\mu} = \frac{1}{n} T_1$$

is an unbiased estimator of μ and hence is an UMVUE of μ. Recall that the same empirical-mean estimator is also an UMVUE of μ when the variance σ^2 is known, and the varaince of the estimator is given by

$$\mathrm{Var}_\theta[\hat{\mu}] = \frac{\sigma^2}{n}$$

which decreases linearly with the sample size n. To find an UMVUE for σ^2, recall that when the mean parameter μ is the known, an UMVUE is given by the empirical variance.

$$x$$

if μ is unknown, it is very tempting to replace μ by its UMVUE $\hat{\mu}$ and consider the estimator

$$\hat{\sigma^2}' = \frac{1}{n} \sum_{i=1}^{n} (X_i - \hat{\mu})^2$$

where

$$\hat{\mu} = \frac{1}{n} \sum j = 1^n X_j$$

Let $X_i = \sigma Z_i + \mu$, where $(Z_1, Z_2, \dots, Z_n)$ are i.i.d. standard Gaussian varaibles. We can write

$$\hat{\sigma^2}' = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \frac{1}{n}\sum_{j=1}^{n}X_j\right)^2$$
$$= \frac{\sigma^2}{n}\sum_{i=1}^{n}\left(Z_i - \frac{1}{n}\sum_{j=1}^{n}Z_j\right)^2$$

now $\mathbb{E}_\theta[\hat{\sigma^2}'] = \frac{\sigma^2}{n}(n-1) = \frac{n-1}{n}\sigma^2$. This estimator is biased for σ^2, but we can get rid of this by multiplying $\hat{\sigma^2}'$ with the scaling factor $(n-1)/n$.

Note that $\sum_{i=1}^{n}(X_i - \hat{\mu})^2 = T_2 - 2(\frac{T_1}{n})T_1 + n(\frac{T_1}{n})^2$ which is a function of $T = (T_1, T_2)$.

The variance of $\hat{\sigma^2}''$ is calculated by

$$\mathrm{Var}_\theta[\hat{\sigma^2}''] = ...$$

This is the price to pay for not knowing the value of the mean parameter μ.

When $n = 1$, X itself is a complete sufficient statistic. Therefore, if an unbiased estimator exists, it is also an UMVUE. As an exercise, show that for $n = 1$, an unbiased estimator for σ^2 does not exist (when both μ and σ^2 are unknown).

Need to find universal delta function such that $\mathbb{E}_\theta[\delta(x)] = \sigma^2\ \forall(\mu, \sigma^2) = \int_{-\infty}^{\infty}\delta(x)\delta_\theta(x)\,dx = \sigma^2$.

5.1.1 Uniform with unknown support $X_i \sim \mathcal{U}([0,\theta])$

Recall that in this case,

$$T = \max_{i \in [n]} X_i$$

is a complete statistic for θ. Start by checking $\mathbb{E}_\theta[T] =?$. $T \leq \theta$.

Recall that the pdf of T can be calculated as

$$f_\theta(t) = nt^{n-1}\theta^{-n}1_{\{0 \leq t \leq \theta\}}$$

The expectation of T is thus given by

$$\mathbb{E}_\theta[T] = \int_0^\theta t f_\theta(t)dt = \frac{n\theta}{n-1}$$

The variance

$$\begin{aligned}\mathrm{Var}_\theta[T] &= \mathbb{E}_\theta[T^2] - (\mathbb{E}_\theta[T])^2 \\ &= \frac{n\theta^2}{n+1} - \left(\frac{n\theta}{n+1}\right)^2\end{aligned}$$

From here we can also compute the varnace of ...

Chapter 6

UMVUE II

6.0.1 UMVUE without a complete sufficient statistic

Rao-Blackwellization can be used to improve the variance of an unbiased estimator. Applying Rao-Blackwellization with a complete sufficient statistic leads to an UMVUE. What if a complete sufficient statistic does NOT exist?

In the previous cases, the model parameter is $\Theta = (0, \infty)$. Here, we shall assume that $\Theta = (1, \infty)$, i.e., we have the additional knowledge that the unknown parameter $\theta > 1$. Note that with the additional knowledge $\theta > 1$, it does not make sense for the estimator to take any value smaller than 1. It is natural to consider a modified estimator

$$\hat{\theta}' = \begin{cases} 1 & , \quad \text{if } T \leq 1 \\ \frac{n+1}{n} T & , \quad \text{if } T > 1 \end{cases}$$

For any $\theta > 1$, the expected value of $\hat{\theta}'$ can be calculated as

$$\begin{aligned} \mathbb{E}_\theta[\hat{\theta}'] &= \int_0^1 f_\theta(t)\, dt + \int_1^0 \frac{n+1}{n} t f_\theta(t)\, dt \\ &= n\theta^{-n} \int_0^1 t^{n-1}\, dt + (n+1)\theta^{-n} \int_1^0 t^n\, dt \\ &= \theta^{-n} + \theta^{-n}(\theta^{n+1} + 1) \end{aligned}$$

Therefore, if T is again a complete sufficient statistic in this case, then $\hat{\theta}'$ is an UMVUE of θ, knowing that $\theta > 1$. Unfortunately, in this case, T is no longer complete, as it can be seen as follows:

Let $\phi(T)$ be a function of T such that $\mathbb{E}_\theta[\phi(T)] = 0,\ \forall \theta > 1$. As before, it can be shown that we must have $\phi(T) = 0,\ \forall t > 1$. However, for $t \leq 1$, $\phi(t)$ does not have to be a zero function to satistfy $\mathbb{E}_\theta[\phi(T)] = 0,\ \forall \theta > 1$.

For example, if we let

$$\phi(t) = \begin{cases} t - \frac{n}{n+1}, & \text{if } t \leq 1 \\ 0, & \text{if } t > 1 \end{cases}$$

Apparently, we have $\mathbb{P}_\theta[\phi(T) = 0] < 1$,

6.0.2 Variance reduction via unbiased estimators of 0

Mathematically, for a given likelihood family $f_\theta(x)$, a statistic $U = U(X)$ is called an unbiased estimator of 0 if $\mathbb{E}_\theta[U] = 0$ for all $\theta \in \Theta$.

Chapter 7

UMVUE III

Feb. 5

Assume that our likelihood function $f_\theta(x)$ satisfies the following two regularity assumptions:

1. The support $\{x : f_\theta(x) > 0\}$ does not depend on θ
2. For any scalar statistic $T(X)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$,

$$\nabla_\theta(\mathbb{E}_\theta[T(X)] = \int_{\mathcal{X}} T(x)\nabla_\theta(f_\theta(x))\, dx$$

Under the above regularity assumptions, the function

$$s_\theta(x) := \nabla_\theta(\log f_\theta(x))$$

is known as the (Fisher) score function. The fundamental identity of score function

$$\mathbb{E}_\theta[s_\theta(X)T(X)] = \nabla_\theta(\mathbb{E}_\theta[T(X)])$$

for any scalar statistic $T(X)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$. Proof of this:

$$\begin{aligned}
\mathbb{E}_\theta[s_\theta(X)T(X)] &= \int_{\mathcal{X}} s_\theta(x)T(x)f_\theta(x)\, dx \\
&= \int_{\mathcal{X}} \nabla_\theta(\log f_\theta(x))T(x)f_\theta(x)\, dx \\
&= \int_{\mathcal{X}} \nabla_\theta(f_\theta(x))T(x)\, dx \\
&= \nabla_\theta\left(\int_{\mathcal{X}} f_\theta(x)T(x)\, dx\right) \\
&= \nabla_\theta(E_\theta[T(X)])
\end{aligned}$$

Applying the score identity with $T(x) = 1$ gives

$$\mathbb{E}_\theta[s_\theta(X)] = \nabla_\theta(\mathbb{E}_\theta[1]) = 0$$

i.e., the score function $s_\theta(X)$ always has a zero mean. The covarience matrix of the score function

$$\mathrm{Var}_\theta[s_\theta(X)] = \mathbb{E}_\theta[s_\theta(X)s_\theta^t(X)]$$

is known as the Fisher information matrix or, in short, Fisher information, and is usually denoted by $I(\theta)$. If the log-likelihood $\log f_\theta(x)$ is twice differentiable with respect to θ and under the additional regularity assumption that for any scalar statistic $T(x)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$,

$$\nabla_\theta^2(\mathbb{E}_\theta[T(x)]) = \int_{\mathcal{X}} T(x)\nabla_\theta^2(f_\theta(x))\,dx$$

We have

$$\nabla_\theta^2(\log f_\theta(x)) = \frac{\nabla_\theta^2(f_\theta(x))}{f_\theta(x)} - s_\theta(x)s_\theta^t(x)$$

and hence, by taking expectation of both sides,

$$I(\theta) = -\mathbb{E}_\theta[\nabla_\theta^2(\log f_\theta(x))]$$

Note that $-\frac{\partial^2}{\partial\theta^2}\log f_\theta(x)$ is the curvature of the log-likelihood $\log f_\theta(x)$ with respect to θ for a given x, so Fisher information measures the average curavature of the log-likelihood $\log f_\theta(x)$ at a given value of θ.

7.1 Re-parameterization

Sometimes the likelihood function $f_\theta(x)$ can be re-parameterized in terms of a more natural parameter

$$\eta = \eta(\theta)$$

In this case, it is usually much easier to first work with the re-parameterized model $\tilde{f}_\eta(x)$ and then translate the result back to the original model $f_\theta(\theta)$.

More specifically, let $s_\eta(x)$ and $I(\eta)$ be Fisher score and the Fisher information with respect to the parameter η, respectively. If $\eta = \eta(\theta)$ is a continuous and differentiable mapping from θ to η, we have

$$s_\theta(x) = \nabla_\theta\left(\log \tilde{f}_\eta(x)\right) = J_\eta^t \nabla_\eta\left(\log \tilde{f}_\eta(x)\right) = J_\eta^t s_\eta(x)$$

where $J_\eta = \left[\frac{\partial\eta_i}{\partial\theta_j}\right]$ is the Jacobian of $\eta = \eta(\theta)$.

7.2 Exponential family of distributions

In particular, an exponential family of distributiosn can be re-parameterized in terms of the natural parameter η as:

$$\tilde{f}_\eta(x) = h(x)\exp(\eta^t T(x) - A(\eta))$$

Note that the score function with respect to the natural parameter can be calculated as:

$$s_\eta(x) = \nabla_\eta \log \tilde{f}_\eta(x) = T(x) - \nabla_\eta A(\eta)$$
$$\text{and } \nabla_\eta^2 \log \tilde{f}_\eta(x) = -\nabla_\eta^2 A(\eta)$$

The Fisher information with respect to η is thus given by

$$I(\eta) = -\mathbb{E}_\eta\left[\nabla_\eta^2 \log \tilde{f}_\eta(x)\right] = \nabla_\eta^2 A(\eta)$$

By the re-parameterization formula, the Fisher information with respect to θ is given by

$$I(\theta) = J_\eta^t I(\eta) J_\eta = J_\eta^t \nabla_\eta^2 A(\eta) J_\eta$$

Additivity over independent measurements

Let $X = (X_1, \ldots, X_n)$ be an observation of n independent measurements:

$$f_\theta(x) = \prod_{i=1}^n f_\theta(x_i)$$

It follows that the score function

$$s_\theta(x) = \sum_{i=1}^n s_\theta(x_i)$$

By the independence of the measurements, the Fisher information

$$I_n(\theta) = \mathrm{Var}_\theta[s_\theta(X)] = \sum_{i=1}^n \mathrm{Var}_\theta[s_\theta(X_i)] = nI_1(\theta)$$

i.e., Fisher information is additive over independent measurements.

7.3 Cramer-Rao lower bound (CRLB)

Assume that the likelihood function $f_\theta(x)$ satisfies the aforementioned regularity assumptions. We have

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq J_\phi I^{-1}(\theta) J_\phi^t, \quad \forall \theta \in \Theta$$

for any estimator $\hat{g}(X)$ of $g(\theta)$ where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)],$$

and $J_\phi = \left[\frac{\partial \phi}{\partial \theta_j}\right]$ is the Jacobian of $\phi = \phi(\theta)$. In particular, if $\hat{g}(X)$ is unbiased, we have $\phi(\theta) = g(\theta)$ and hence

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq J_g I^{-1}(\theta) J_g^t, \quad \forall \theta \in \Theta$$

Therefore, any unbiased estimator whose variance equals

$$J_g I^{-1}(\theta) J_g^t$$

is the UMVUE for $g(\theta)$.

Proof. Let us assume that θ is a scalar parameter. The vector case follows essentially the same idea, just requiring more machinery. By the Cauchy-Schwarz inequality, we have

$$\mathrm{Var}_\theta[s_\theta(X)] \, \mathrm{Var}_\theta[\hat{g}(X)] \geq (\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)])^2$$

so

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \frac{(\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)])^2}{\mathrm{Var}_\theta[s_\theta(X)]}$$

Recall that

$$\mathbb{E}_\theta[s_\theta(X)] = 0, \quad \text{and} \quad \mathrm{Var}_\theta[s_\theta(X)] = I(\theta)$$

We thus have

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \frac{(\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)])^2}{I(\theta)}$$

By the score identity, the covariance

$$\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)] = \mathbb{E}_\theta[s_\theta(X)\hat{g}(X)] - \mathbb{E}_\theta[s_\theta(X)]\mathbb{E}_\theta[\hat{g}(X)] = \frac{d}{d\theta}\mathbb{E}_\theta[\hat{g}(X)] = \frac{d}{d\theta}\phi(\theta)$$

It follows that

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \frac{(\frac{d}{d\theta}g(\theta))^2}{I(\theta)}$$

for all $\theta \in \Theta$. □

7.4 Efficient estimators

In statistics, we say $\hat{g}(X)$ is an efficient estimator of $g(\theta)$ if it is unbiased and its variance equals the CRLB. By definition, an efficient estiamator, if exists, must be the UMVUE. However, the variance of the UMVUE may be strictly greater than the CRLB. In this case, an efficient estimator does not exist. Keep in mind that Fisher information and, by extension, CRLB are only defined for likelihood families that satistify the regularity conditions.

7.4.1 Hammersley-Chapman-Robbins lower bound

We have

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{(\phi(\theta') - \phi(\theta))^2}{\mathbb{E}_\theta\left[(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1)^2\right]}, \quad \forall \theta \in \Theta$$

for any estiamator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)]$$

In particular, if $\hat{g}(X)$ is unbiased, we have $\phi(\theta) = g(\theta)$ and hence

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{(g(\theta') - g(\theta))^2}{\mathbb{E}_\theta\left[(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1)^2\right]}, \quad \forall \theta \in \Theta$$

Let $\theta' = \theta + \epsilon$. Under the regularity conditions, it can be shown that

$$\lim_{||\epsilon|| \to 0} \frac{(\phi(\theta') - \phi(\theta))^2}{\mathbb{E}_\theta\left[(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1)^2\right]} = J_\phi I^{-1}(\theta) J_\phi^t, \quad \forall \theta \in \Theta$$

Therefore, the HCRLB is at least as tight as the CRLB under the regularity conditions. The real power of the HCRLB, however, lies in the fact that it can be applied even when the regularity conditions are not met.

Example 7.4.1. Uniform with unknown support ($\theta > 0$). Here the support of $f_\theta(x)$ depends on θ. Therefore, the regularity conditions are not met for this likelihood family and the CRLB does not apply. In fact, CRLB for any observation consists of n i.i.d. measurements decreases linearly with the sample size n.

However, as shown before, the variance of the UMVUE in this case decreases quadratically with the sample size n. To overcome this issue, we can choose $\theta' > \theta$‘ in the evaluation of the HCRLB, so the support of $f_\theta(x)$ is a subset of the support of $f_{\theta'}(x)$. This gives

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' > \theta} \frac{(\theta' - \theta)^2}{\left(\frac{\theta'}{\theta}\right)^n - 1} = \sup_{\epsilon > 0} \frac{\theta^2 \epsilon^2}{(1+\epsilon)^n - 1}$$

7.5 End discussion

$$\theta = \exp(-\lambda) = \mathbb{P}(X_i = 0) = \exp(-1\lambda)$$

The only unbiased estimator is $T(X) = (-1)^X$, so estimate is either -1 or 1 depending on the parity of X.

We know that θ can only be positive. Practically, only considering unbiased estimator can have negative consequences.

Chapter 8

Maximum-Likelihood Estimation

Feb. 10

Finding an UMVUE can be mathematically challenging, and in some cases, it might not even exist. More importantly, while an UMVUE minimizes variance among unbiased estimators, strictly adhering to the unbiased constraint can somestimes lead to a significantly higher variance than a slightly biased estimator.

Illuminating examples

Perhaps the best known example

$$X \sim Poisson(\lambda),$$

where $\lambda > 0$ is the unknown parameter. If our goal is to estimate

$$g(\theta) = \exp(-2\lambda),$$

then it can be shown that the only unbiased estimator is give by the laughable

$$\hat{g}(X) = (-1)^X$$

Another illuminating example is to estimate the location of a light souce based on its projection on the boundary of a dart board over a randomly placed dart.

8.1 Maximum-likelihood estimator (MLE)

An alternative frequentist estimator is the maximum-likelihood estimator (MLE). Focusing on estimating the model parameter θ itself. Given a likelihood family $f_\theta(x)$, an estimator $\hat{\theta} = \hat{\theta}(X)$ is an MLE of θ if

$$\hat{\theta}(x) \in \arg\max_{\theta\in\Theta} f_\theta(x), \quad \forall x \in \mathcal{X}$$

or, equivalently,

$$\hat{\theta}(x) \in \arg\max_{\theta\in\Theta} \log f_\theta(x), \quad \forall x \in \mathcal{X}$$

8.1.1 Maximizing likelihood and minimizing K-L divergence

While mainly motivated by intuition, MLE has an interesting interpretation when the observation $X = (X_1, \ldots, X_n)$, where

$$X_i \overset{iid}{\sim} p_\theta(x_i)$$

More specifically, given an observation $x = (x_1, \ldots, x_n)$, let

$$\hat{p}_x(z) = \frac{1}{n}\sum_{i=1}^{n} 1_{\{x_i=z\}}, \quad \forall z \in \mathcal{X}$$

be the *empirical* estimate of the *marginal* distribution given the observation x. For two pmf's p and q over the same alphabet $\mathcal{X}$, the **K-L divergence** between p and q is defined as:

$$D_{KL}(p||q) := \mathbb{E}_p\left[\log\frac{p(Z)}{q(Z)}\right] = \sum_{z\in\mathcal{X}} p(z)\log\frac{p(z)}{q(z)}$$

Information inequality

Occurs when

$$D_{KL}(p||q) \geq 0$$

where the equality holds if and only if $p = q$. In statistics, information theory, and machine learning, one often use K-L divergence as a distance measure between two probability distributions, even though this distance measure is generally asymmetric about the two input distributions.

Claim: maximizing the likelihood

$$\sum_{i=1}^{n} \log p_\theta(x_i)$$

is equivalent to minimzing the K-L divergence

$$D_{KL}(\hat{p}_x||p_\theta)$$

Proof. First note that

$$\begin{aligned} D_{KL}(\hat{p}_x||p_\theta) &= \mathbb{E}_{\hat{p}_x}\left[\log\frac{\hat{p}_x(Z)}{p_\theta(Z)}\right] \\ &= \mathbb{E}_{\hat{p}_x}[\log \hat{p}_x(Z)] - \mathbb{E}_{\hat{p}_x}[\log p_\theta(Z)] \end{aligned}$$

Further note that

$$\begin{aligned} \mathbb{E}_{\hat{p}_x} &= \sum_{z\in\mathcal{X}} \hat{p}_x(z)\log p_\theta(z) \\ &= \frac{1}{n}\sum_{z\in\mathcal{X}}\sum_{i=1}^{n} 1_{\{x_i=z\}}\log p_\theta(z) \\ &= \frac{1}{n}\sum_{i=1}^{n}\left(\sum_{z\in\mathcal{X}} 1_{\{x_i=z\}}\log p_\theta(z)\right) \\ &= \frac{1}{n}\sum_{i=1}^{n}\log p_\theta(x_i) \end{aligned}$$

Combining the above two results gives

$$D_{KL}(\hat{p}_x||p_\theta) = \mathbb{E}_{\hat{p}_x}[\log \hat{p}_x(Z)] - \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(x_i)$$

We may thus conclude that minimizing the K-L divergence $D_{KL}(\hat{p}_x||p_\theta)$ is equivalent to maximizing the likelihood $\sum_{i=1}^{n} \log p_\theta(x_i)$. The above result also suggests that if our model is identifiable, MLE will recover θ in the limit as $n \to \infty$. We will have more to say about this result a bit later. □

8.1.2 Re-parameterization

Assume that the likelihood function $f_\theta(x)$ can be re-parameterized in terms of a more natural parameter

$$\eta = \eta(\theta)$$

In this case, for any given observation x, we have

$$\arg\max_{\theta\in\Theta} f_\theta(x) = \{\theta \in \Theta : \eta(\theta) \in \arg\max_{\theta\in\Theta}\}$$

... Consider an exponential family of distribution re-parameterized by the natural parameter η:

$$\tilde{f}_\eta(x) = h(x)\exp(\eta^t T(x) - A(\eta))$$

where the space of the natural parameter is an open and connected subset of $\mathbb{R}^s$. To find an MLE for η, note that

$$\begin{aligned}
\nabla_\eta \log \tilde{f}_\eta(x) &= T(x) - \nabla_\eta A(\eta) = T(x) - \mathbb{E}_\eta[T(X)] \\
\nabla_\eta^2 \log \tilde{f}_\eta(x) &= -\nabla_\eta^2 A(\eta) = \mathrm{Cov}_\eta[T(X)]
\end{aligned}$$

A covariance mastrix is always positive semi-definite, so for any $x \in \mathcal{X}$, the log-likelihood $\log \tilde{f}_\eta(x)$ is a concave function of η, for which any stationary point is a global maxima. Therefore, any solution for η to the equation

$$\mathbb{E}_\eta[T(X)] = T(x)$$

is an MLE for η. Furthermore, if the covariance matrix of $T(X)$ is positive definite for all η, for any $x \in \mathcal{X}$ the log-likelihood $\log \tilde{f}_\eta(x)$ is a strictly concave function of η. In this case, the MLE is unique if it exists.

Finally, by the re-parameterization property, any solution for θ to the equation

$$\mathbb{E}_\theta[T(X)] = T(x)$$

is an MLE for θ.

Gaussian with unknown mean and variance

Recall that in this case, a sufficient statistic is given by

$$T_1 = \sum_{i=1}^{n} X_i \quad \text{and} \quad T_2 = \sum_{i=1}^{n} X_i^2$$

the expectations of the sufficient statistics are given by:

$$\mathbb{E}_\theta[T_1] = n\mu \quad \text{and} \quad \mathbb{E}_\theta[T_2] = n(\sigma^2 + \nu^2)$$

From previous discussion, any solution of (μ, σ^2) to the equations

$$\begin{aligned} n\mu &= T_1(x) \\ n(\sigma^2 + \mu^2) &= T_2(x) \end{aligned}$$

is an MLE. The above equations have a unique solution given by

$$\mu = \frac{T_1(x)}{n} \quad \text{and} \quad \sigma^2 = \frac{T_2(x)}{n} - \left(\frac{T_1(x)}{n}\right)^2$$

We may thus conclue that the empirical mean

$$\hat{\mu} = \frac{T_1(x)}{n} = \frac{1}{n}\sum_{i=1}^n X_i$$

is the MLE for μ and the empirical variance

$$\hat{\sigma}^2 = \frac{T_2(x)}{n} - \left(\frac{T_1(x)}{n}\right)^2 = \frac{1}{n}\sum_{i=1}^n X_i^2 - \hat{\mu}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \hat{\mu})^2$$

is the MLE for σ^2, when both μ and σ^2 are unknown. By Cochran's theorem,

$$\mathbb{E}_\theta\left[\sum_{i=1}^n (X_i - \hat{\mu})^2\right] = (n-1)\sigma^2$$

so

$$\mathbb{E}_\theta[\hat{\sigma^2}] = \frac{n-1}{n}\sigma^2$$

i.e., the empirical variance is a biased estimator for the true variance σ^2.

8.1.3 Consistency and asymptotic normality

Note that in the previous example, while the MLE for σ^2 is biased, the bias vanishes asymptotically in the limit as the sample size $n \to \infty$. Also note that the variances of the MLE for both μ and σ^2 coincide with the CRLB for unbiased estimators. Under certain regularity conditions, it can be shown that the above observations hold for a general likelihood function.

More specifically, let $X = (X_1, \ldots, X_n)$, where $X_i \overset{iid}{\sim} f_\theta(x_i)$. For any θ that is in the interior of Θ, let $\hat{\theta}$ be the MLE for θ given by the unique solution to the equation

$$\nabla_\theta\left(\sum_{i=1}^n \log f_\theta(x_i)\right) = 0$$

We have $\sqrt{n}(\hat{\theta} - \theta)$ converges in distribution to $\mathcal{N}(0, I_1^{-1}(\theta))$ in the limit as the sample size $n \to \infty$.

That is, when the sample size n is large, the MLE $\hat{\theta}$ is approximately distributed as

$$\mathcal{N}\left(\theta, \frac{1}{n} I_1^{-1}(\theta)\right)$$

or, equivalently,

$$\mathcal{N}(\theta, I_n^{-1}(\theta))$$

due to the additivity of Fisher information $I_n(\theta) = nI_1(\theta)$. Therefore, under certain regularity conditions, the MLE is consistent in that it can recover the unknown parameter θ in the limit as the sample size $n \to \infty$. The proof of the asymptotic normality of the MLE is based on the well-known central limit theorem. This is omitted here, however.

8.1.4 Multinomial with unknown probabilities

Goal is to use the empirical counts to estimate the true probability. Let $X \sim p_\theta(x)$, where the observation

$$X = (N_1, \ldots, N_K)$$

is the empirical counts over K categories and n total counts, the unknown parameter

$$\theta = (p_1, \ldots, p_K)$$

is a probability vector over K categories, and the likelihood function is multinomial:

$$p_\theta(x) = \frac{n!}{n_1! \cdots n_K!} \prod_{k=1}^{K} p_k^{n_k}$$

The total counts n and the number of categories K are assumed to be known. The MLE can be found by solving the equation

$$\nabla_\theta L_\lambda(\theta) = 0$$

and then finding a Lagrangian multiplier $\lambda \geq 0$ to satisfy the constraint

$$\sum_{k=1}^{K} p_k = 1$$

The MLE for θ in this case is given by the empirical frequency estimator

$$\hat{\theta} = \left(\frac{N_1}{n}, \ldots, \frac{N_K}{n}\right)$$

Note that for any $k \in [K]$

$$\mathbb{E}_\theta[N_k] = np_k$$

so the empirical frequency estimator $\hat{\theta}$ is unbiased.

8.2 Estimating $g(\theta)$

Let $\eta = g(\theta)$. Note that if the likelihood function $f_\theta(x)$ can be rep-parameterized in terms of η, then it is natural to use an MLE of η as an "MLE" estimate of $g(\theta)$. By the re-parameterization property, η^* is an MLE of η if and only if

$$\eta^* = g(\theta^*)$$

for some θ^* that is an MLE of θ. Therefore, if the likelihood function $f_\theta(x)$ can be re-parameterized in terms of $g(\theta)$, the "MLE" of $g(\theta)$ is given by $g(\theta^*)$, where θ^* is an MLE of θ.

In statistics, it is customary to define the MLE of $g(\theta)$ as $g(\theta^*)$, whether the likelihood function $f_\theta(x)$ can be re-parameterized in terms of $g(\theta)$ or not. This is known as the invariance property of MLE. Many other estimators such as UMVUE do not enjoy the invariance property.

Chapter 9

Expectation-Maximization

Feb. 12

In general, finding MLE requires iterative numerical techniques. Gradient descent methods for minimization as Newton-Raphson and Fisher's scoring methods are commonly used. Here we discuss a method known as expectation-maximization (EM) algorithm, which is specifically designed for computing MLE for models with latent variables.

9.1 Models with latent variables

Let $(X, U) \sim f_\theta(x, u)$, where X is the observation and U is the unobserved/latent variable. We seek the ML estimate of θ based on the observation X:

$$\hat{\theta}(x) \in \arg\max_{\theta \in \Theta} \log f_\theta(x)$$

where

$$f_\theta(x) = \int_{\mathcal{U}} f_\theta(x, u)\, du$$

The above obtimization problem can be difficult to solve analytically due to the marginalization of the latent variables, which occurs inside the logarithm of the objective function.

9.1.1 Mixture Gaussian model

Let $X = (X_1, \ldots, X_n)$ and $U = (U_1, \ldots, U_n)$, where (X_i, U_i) are i.i.d. according to:

$$U_i \sim Cat(u_i; \pi_1, \ldots, \pi_K) \quad \text{and} \quad X_i | U_i \sim \mathcal{N}(x_i; \mu_{i_i}, \sigma^2_{u_i})$$

Marginalizing over U_i gives

$$f_\theta(x_i) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2)$$

which is a mixture Gaussian distribution, and

$$\theta = (\pi_1, \ldots, \pi_k, \mu_1, \ldots, \mu_K, \sigma_1^2, \ldots, \sigma_K^2)$$

is the unknown parameter. The log-likelihood for all measurements is given by

$$\log f_\theta(x) = \sum_{i=1}^{n} \log f_\theta(x_i) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2) \right)$$

Finding MLE for $(\mu_1, \dots, \mu_K)$

Assume for the moment that $(\pi_1, \dots, \pi_K)$ and $(\sigma_1^2, \dots, \sigma_K^2)$ are known. To find the MLE for $(\mu_1, \dots \mu_K)$, note that for any $j \in [K]$ we have

$$\frac{\partial}{\partial \mu_j} \log f_\theta(x) = \sum_{i=1}^n \frac{\tilde{\pi}_j(x_i)(x_i - \mu_j)}{\sigma_j^2}$$

where

$$\tilde{\pi}_j(x_i) = \frac{x_i \mathcal{N}(x_i; \mu_j, \sigma_j^2}{\sum_{k=1}^K x_i \mathcal{N}(x_i; \mu_k, \sigma_k^2)} = \mathbb{P}_\theta(U_i = j | X_i = x_i)$$

Setting the partial derivatives equal to zero for all j gives the following set of equations:

$$\sum_{i=1}^n \frac{\tilde{\pi}_j(x_i)(x_i - \mu_j)}{\sigma_j^2} = 0, \quad \forall j \in [K]$$

for which the solution for $(\mu_1, \dots, \mu_K)$ is the MLE.

A vicious circle

Note that if $\tilde{\pi}_j(x_i)$'s are known, the above equations are separate for each μ_j and hence are very easy to solve:

$$\mu_j = \frac{\sum_{i=1}^n \tilde{\pi}_j(x_i) x_i}{\sum_{i=1}^n \tilde{\pi}_j(x_i)}, \quad \forall j \in [K]$$

However, in reality, $\tilde{\pi}_j(x_i)$'s depend on $(\mu_1, \dots, \mu_K)$ and hence are unknown. Therefore, the above equations are interlocked and hence are very difficult to solve. This is a common challenge for finding the MLE for models with latent variables. We are now in the following situation:

1. If we know $\tilde{\pi}_j(x_i)$'s, we can solve for μ_j's
2. If we know μ_j's, we can compute $\tilde{\pi}_j(x_i)$'s

9.2 An iterative procedure

A successive approximation procedure:

1. Initialize $(\mu_1, \dots, \mu_K)$
2. Evaluate the posterior probabilities $(\tilde{\pi}_1(x_i), \dots, \tilde{\pi}_K(x_i)), i \in [n]$ using the current value of $(\mu_1, \dots, \mu_K)$
3. Update the value for $(\mu_1, \dots, \mu_K)$ using the posterior probabilities from Step 2
4. Stop if the value for $(\mu_1, \dots, \mu_K)$ has already converges; otherwise go back to Step 2

Next, we shall generalize the above prodedure to general likelihood models with latent variables, and the resulting procedure is known as the expectation-maximization (EM) algorithm.

9.3 A variational characterization of the log-likelihood

The main idea behind the EM algorithm is the following variational characterization of the log-likelihood:

$$\log f_\theta(x) = \max_q F(q, \theta)$$

where

$$F(q,\theta) = \log f_\theta(x) - D_{KL}(q(u)||f_\theta(u|x)) \tag{9.1}$$
$$= \mathbb{E}_q[\log f_\theta(x, U)] - \mathbb{E}_q[\log q(U)], \tag{9.2}$$

and both expectations are over $U \sim q(u)$. The main advantage of this variational characterization is that it allows us to rewrite the problem of finding the MLE as a double-maximization problem:

$$(\hat{\theta}, q^*) \in \arg\max_{(\theta,q)} F(q, \theta)$$

Proof. First note that by the information inequality, for any density function q we have

$$D_{KL}(q(u)||f_\theta(u|x)) \geq 0$$

where the equality holds if and only if

$$q(u) = f_\theta(u|x), \quad \forall u \in \mathcal{U}$$

We thus have

$$\log f_\theta(x) \geq \log f_\theta(x) - D_{KL}(q(u)||f_\theta(u|x))$$

where the equality holds if and only if

$$q(u) = f_\theta(u|x), \quad \forall u \in \mathcal{U}$$

and hence

$$\log f_\theta(x) = \max_q(\log f_\theta(x) - D_{KL}(q(u)||f_\theta(u|x)))$$

Further note that

$$\begin{aligned}
&\log f_\theta(x) - D_{KL}(q(u)||f_\theta(u|x)) \\
&\quad = \log f_\theta(x) - (\mathbb{E}_q[\log q(U)] - \mathbb{E}_q[\log f_\theta(U|x)]) \\
&\quad = \mathbb{E}_q[\log f_\theta(x)] + \mathbb{E}_q[\log f_\theta(U|x)] - \mathbb{E}_q[\log q(U)] \\
&\quad = \mathbb{E}_q[\log(f_\theta(x) f_\theta(U|x))] - \mathbb{E}_q[\log q(U)] \\
&\quad = \mathbb{E}_q[\log f_\theta(x, U)] - \mathbb{E}_q[\log q(U)]
\end{aligned}$$

$\square$

We have thus proved that (2.1) and (2.2) are in fact equivalent to each other.

9.4 Finding MLE via alternating maximization

Recall that the MLE can be found by solving the double-maximization problem:

$$\max_{(\theta,q)} F(q,\theta)$$

Consider the following alternating-maximization strategy for solving the above double-maximization problem:

1. Initialize $\theta = \theta^{(0)}$ and $t = 1$
2. Repeat the following steps until the values of (θ, q) converge:
   (a) Fix $\theta = \theta^{(t-1)}$ and find
   $$q^{(t)} \in \arg\max_q F\left(q, \theta^{(t-1)}\right)$$
   (b) Fix $q = q^{(t)}$ and find
   $$\theta^{(t)} = \arg\max_\theta F\left(q^{(t)}, \theta\right)$$
   (c) $t \leftarrow t + 1$

The problem of fixing $\theta = \theta^{(t-1)}$ and finding

$$q^{(t)} \in \arg\max_q F\left(q, \theta^{(t-1)}\right)$$

is trivial. By the information inequality,

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x), \quad \forall u \in \mathcal{U}$$

which gives

$$F\left(q^{(t)}, \theta^{(t-1)}\right) = \log f_{\theta^{(t-1)}}(x)$$

By comparison, the problem of fixing $q = q^{(t)}$ and finding

$$\theta^{(t)} \in \arg\max_\theta F\left(q^{(t)}, \theta\right)$$

is much more involved. To solve this optimization problem, we shall use expression (2) and write

$$F\left(q^{(t)}, \theta\right) = \mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)] - \mathbb{E}_{q^{(t)}}[\log q^{(t)}(U)]$$

Note that the second term in the above expression does *not* involve θ. Thus, $\theta^{(t)}$ can be found by maximizing over the first term in the above expression

$$\theta^{(t)} \in \arg\max_\theta \mathbb{E}_{q^{(t)}}[\log f_\theta(x, U)]$$

where

$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x), \quad \forall u \in \mathcal{U}$$

The above two equations allow us to continuously update the value of (θ, q), leading to the famous EM algorithm. **Feb. 17**†

Chapter 10

Generative Adversarial Networks

Feb. 19

Generative adversarial network (GAN) is a powerful mathematical framework for learning a generative model. There are two distinct features for GAN. First, GAN uses a deterministic generator $g_\theta(z)$ to produce new data samples. Second, GAN uses the Jensen-Shannon (JS) divergence to measure the distance between the target data-generating distribution $p_\theta(x)$ and the true data-generating distribution $p(x)$:

$$D_{JS}(p_\theta \| p) = \frac{D_{KL}(p_\theta \| \frac{p_\theta + p}{2}) + D_{KL}(p \| \frac{p_\theta + p}{2})}{2}$$

where

$$D_{KL}(p_\theta \| p) = \mathbb{E}_\theta \left[\log \frac{p_\theta}{p} \right]$$

Note that unlike the K-L divergence, the J-S divergence is symmetric between its two input distributions.

The main challenge for GAN is that the J-S divergence $D_{JS}(p_\theta \| p)$ is very difficult to estimate from the training data examples. One way to address this challenge is through the following variational characterization of the J-S divergence:

$$D_{JS}(p_\theta \| p) = \max_d \frac{\mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1 - d(X))] + 2\log 2}{2}$$

where the maximization is over a discriminator $d : \mathcal{X} \to [0, 1]$. An immediate consequence of the above variational characterization is that the new objective

$$L(\theta, d) := \mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1 - d(X))]$$

can now be easily estimated from the training data examples

$$L(\theta, d) \approx \frac{1}{m'} \sum_{i=1}^{m'} \log d(g_\theta(z_i)) + \frac{1}{m} \sum_{i=1}^{m} \log(1 - d(x_i))$$

where z_i's are i.i.d. samples of the random source Z.

The problem of learning a generator can thus be posed as a min-max optimization problem:

$$\min_{\theta} \max_{d} \left(\frac{1}{m'} \sum_{i=1}^{m'} \log d(g_\theta(z_i)) + \frac{1}{m} \sum_{i=1}^{m} \log(1 - d(x_i)) \right)$$

Just like the EM algorithm, one may consider an alternating-optimization strategy for solving the above min-max problem.

10.0.1 Proof of the variational characterization

First note that

$$\begin{aligned}
&\mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1 - d(X))] \\
&= \int_{\mathcal{X}} (p_\theta(x) \log d(x) + p(x) \log(1 - d(x)))\, dx
\end{aligned}$$

To maximize the above objective over d, we can choose $d(x) \in [0, 1]$ to maximize

$$p_\theta(x) \log d(x) + p(x) \log(1 - d(x))$$

for each $x \in \mathcal{X}$. Consider for each $x \in \mathcal{X}$, the problem of maximizing

$$g(t) := p_\theta(x) \log t + p(x) \log(1 - t)$$

over $t \in [0, 1]$. Setting $g'(t) = 0$ gives:

$$\frac{p_\theta(x)}{t} - \frac{p(x)}{1 - t} = 0 \quad \Rightarrow \quad t = \frac{p_\theta(x)}{p_\theta(x) + p(x)}$$

Chapter 11

Decision-Theoretic Framework

In a decision-theoretic framework, the performance of an estimator is evaluated through a loss function. Formally, a decision rule $\delta : \mathcal{X} \to \mathcal{D}$ is a mapping from the observation space $\mathcal{X}$ to a decision space $\mathcal{D}$. For parameter estimation, $\mathcal{D} = \Theta$, and for detection $\mathcal{D} = \{0, 1\}$. More generally, however, $\mathcal{D}$ can be the colletion of any viable decisions. A loss function is any function $L : \Theta \times \mathcal{D} \to [0, \infty)$, which is supposed to evaluate the penalty (or error) associated with the decision d when the parameter takes the value θ. The frequentist risk associated with a decision rule δ is defined as

$$R_\delta(\theta) := \mathbb{E}_\theta[L(\theta, \delta(X))]$$

where the expectation is over $X \sim f_\theta(x)$.

11.0.1 Usual loss functions

Where the decisions have a numerical nature, the most usual choice of the loss function is the squared-error loss

$$L(\theta, d) = (\theta - d)^2$$

A similar but less smooth function is the absolute-error loss

$$L(\theta, d) = |\theta - d|$$

Note that both the squared-error loss and the absolute-error loss are convex functions of d for any given θ, so both are examples of a convex loss function. When the decisions have a categorical nature, the most usual choice of the loss function is the 0-1 loss

$$L(\theta, d) = 1_{\{\theta \neq d\}}$$

for parameter estimation problems and

$$L(\theta, d) = 1_{\{\theta \notin \Theta_d\}}$$

for detection.

11.0.2 MSE and bias-variance tradeoff

Under the squared-error loss, the frequentist risk is given by the mean squared-error (MSE), which can be further decomposed as the sum of a bias term and a variance term:

$$\begin{aligned}
R_\delta(\theta) &= \mathbb{E}_\theta[(\theta - \delta(X))^2] \\
&= \mathbb{E}_\theta[(\theta - \mathbb{E}_\theta[\delta(X)] + \mathbb{E}_\theta[\delta(X)] - \delta(X))^2] \\
&= (\theta - \mathbb{E}_\theta[\delta(X)])^2 + \mathbb{E}_\theta[(\mathbb{E}_\theta[\delta(X)] - \delta(X))^2] \\
&= (\theta - \mathbb{E}_\theta[\delta(X)])^2 + \mathrm{Var}_\theta[\delta(X)]
\end{aligned}$$

In short,

$$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance}$$

Potentially, there is a tradeoff between the bias term and the variance terms in achieving a smaller value of the MSE.

Gaussian with unknown mean and variance

Consider the following family of estimators

$$\delta_b(X) = bS^2$$

for the variance parameter σ^2, where

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

is the UMVUE for σ^2, and b can be any positive constant. By Cochran's theorem, we have

$$Bias^2 = (b-1)^2\sigma^4$$

$$Variance = \frac{2b^2\sigma^4}{n-1}$$

and hence

$$MSE = \left((b-1)^2 + \frac{2b^2}{b-1}\right)\sigma^4$$

Note that while for $b \geq 1$, both the bias and the variance terms increase monotonically with b, for $b \in [0,1]$ the bias term decreses while the variance term increase monotonically with b. Therefore, for $b \in [0,1]$ there is a tradeoff between the bias and the variance terms in achieving a smaller value of the MSE. In particular, we can compare the stimators with $b = 1$ (the UMVUE) and $b = \frac{n-1}{n}$ (the MLE): the UMVUE has a smaller bias but a larger variance than the MLE. In terms of the MSE, we have

$$\mathbb{E}_\theta[(\theta - \delta_{\frac{n-1}{n}}(X))^2] = \frac{(2n-1)\sigma^4}{n^2} < \frac{2\sigma^4}{n-1} = \mathbb{E}_\theta[(\theta - \delta_1(X))^2]$$

However, neither achieves the minimum MSE, which is achieved at $b = \frac{n-1}{n+1}$ with the minimum MSE given by

$$\mathbb{E}_\theta[(\theta - \delta_{\frac{n-1}{n+1}}(X))^2] = \frac{2\sigma^4}{n+1}$$

11.0.3 Frequentist risk as an evalutation criterion

Note that for a decision rule δ, the frequentist risk $R_\delta(\theta)$ is a function of the unknown parameter θ. Therefore, the frequentist criterion does not induce a total ordering on the set of decision rules. It is thus generally impossible to compare decision procedures with this criterion, since two crossing risk functions prevent comparison between the corresponding decision rules. One may hope for a decision rule δ that uniformly minimizes the risk $R_\delta(\theta)$, but such cases rarely occur unless the space of decision rules is restricted.

11.0.4 Admissability

Formally, a decision rule δ is inadmissable if it is dominated by another decision rule δ', i.e, for every $\theta \in \Theta$ we have

$$R_{\delta'}(\theta) \leq R_\delta(\theta)$$

and there exists at least one value $\theta_0 \in \Theta$ such that

$$R_{\delta'}(\theta_0) < R_\delta(\theta_0)$$

Otherwise, δ is said to be admissable. So at least in theory, inadmissable estimators should not be considered since they can be uniformly improved. For example, for Gaussian with unknown mean and variance, both the UMVUE $\delta_1 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$ and the MLE $\delta_{\frac{n-1}{n}} = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2$ are inadmissable with respect to the squared-error loss, as both are dominated by $\delta_{\frac{n-1}{n+1}} = \frac{1}{n+1}\sum_{i=1}^n (X_i - \bar{X})^2$.

Chapter 12

Uniformly Minimum-Risk Equivariant Estimation